

Invertible Matrices

Definition

We say an $n \times n$ matrix A is invertible if there is a matrix B such that $AB = BA = I_n$. We call B the inverse of A and write A^{-1} for B .

Fact:

- If A is invertible then A^{-1} is uniquely defined.
- If A is invertible then the associated linear transformation is one-to-one and onto.

Ex:

$$i) \underline{A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}}, \quad \underline{A^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}}.$$

Check: $AA^{-1} = I_2 = A^{-1}A.$

ii) $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \leftarrow \text{not invertible.}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \underline{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}}$$

never
equal to I_2 .

As a trans., $T_A(e_2) = 0$ so T_A is not 1-1.

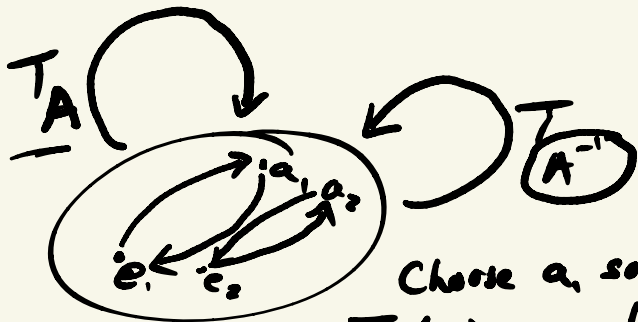
$$\text{iii)} \quad \frac{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}{= \begin{pmatrix} \boxed{ad-bc} & 0 \\ 0 & \boxed{ad-bc} \end{pmatrix}}.$$

$ad-bc \neq 0$ implies that $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible. In fact the inverse is.

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Why if T_A is 1-1 and onto, is A invertible?

Say A is $n \times n$.



Choose a_1 so that $T_A(a_1) = e_1$ (because T_A is onto)

Choose $a_2, \dots, a_n$ s.t. T_A is 1-1

$$A^{-1} = \begin{pmatrix} a_1 & \dots & a_n \\ 1 & & 1 \end{pmatrix} T_A(a_i) = e_i$$

Elementary Matrices

Definition

We call an $n \times n$ matrix E an elementary matrix if E can be obtained from I_n by a single elementary row operation.

Types of elementary matrices

- E could be obtained by exchanging two rows of I_n .
- E could have 1's on the diagonal with possibly one non-zero entry not 1.
- E could have 1's on the diagonal and possibly one non-zero entry somewhere else in the matrix.

Three types of elem. matrices:

i) Exchanging rows:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \leadsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{array}{l} \text{inter.} \\ \text{rows} \\ \text{2 and 3.} \end{array}$$

↑
elem. matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 6 & -1 \\ 2 & 0 & 4 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 6 & -1 \\ 1 & 3 & 2 \\ 2 & 0 & 4 \end{pmatrix}$$

Mult. by the elem. matrix performs the row operation!

Ex:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \leadsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{array}{l} \text{mult.} \\ \text{row 2} \\ \text{by 2.} \end{array}$$

↑
elem. matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 6 & 8 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

Again, mult. by the elem. matrix
performed to oper. on the other matrix.

Ex:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{array}{l} -3 \times R3 \\ + R1. \end{array}$$

↑
elem matrix.

$$\begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 3 \\ 0 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 3 & -4 \\ 1 & 1 & 3 \\ 0 & -1 & 2 \end{pmatrix}$$

Again, the matrix performs the row operation !.

Claim: All elem. matrices are invertible.

$$\textcircled{1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\textcircled{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\textcircled{3} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

How to find A^{-1}

Why elementary matrices?

Every elementary matrix is invertible and if E is elementary then EA is the same as performing the elementary row operation which got E on A .

An algorithm for finding A^{-1}

- 1 Start with the matrix $[A \ I]$.
- 2 Row reduce to reduced row echelon form.
- 3 If the reduced row echelon form is $[I \ B]$ then B is A^{-1} .
- 4 Otherwise A is not invertible.

Ex:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

Start with

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

ref.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boxed{\begin{pmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}}$$

$\uparrow A^{-1}$